

CMP-19 - Privacy and Data Governance Service

Own data classification, purpose, authority, minimization, consent, retention, disclosure, deletion, and inference-risk policy state.

Authority and state ownership

Book I: Decisions 24-29, 35, 40-43, 47-50, and 58. Book II: Chapters 19. Authoritative state: Data Classification Records, purpose bindings, retention classes, consent and authority records, deletion obligations. Explicitly non-owned: source data contents, authoritative Identity state, Treaty lifecycle, application business logic.

Critical invariant

Data use is permitted only for declared authorized purpose with minimization, retention, disclosure, and disposal controls.

Logical interfaces

CMP-19-IF-01 / Command / Classify Data

Command `Classify Data` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-19-IF-02 / Command / Authorize Purpose

Command `Authorize Purpose` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-19-IF-03 / Command / Set Retention

Command `Set Retention` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-19-IF-04 / Command / Order Deletion

Command `Order Deletion` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-19-IF-05 / Command / Restrict Disclosure

Command `Restrict Disclosure` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-19-IF-06 / Query / Get Data Policy

Query `Get Data Policy` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-19-IF-07 / Query / Evaluate Data Use

Query `Evaluate Data Use` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-19-IF-08 / Query / Get Deletion Status

Query `Get Deletion Status` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-19-IF-09 / Event / Data Classified

Event `Data Classified` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-19-IF-10 / Event / Purpose Authorized

Event `Purpose Authorized` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-19-IF-11 / Event / Deletion Completed

Event `Deletion Completed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-19-IF-12 / Event / Use Denied

Event `Use Denied` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Unclassified > Classified > Active > Restricted > Disposed. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: purpose drift, overcollection, retention breach, inference leakage, deletion incompleteness. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-19-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-19-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-19-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-19-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-19-TST-005 - Critical invariant. Data use is permitted only for declared authorized purpose with minimization, retention, disclosure, and disposal controls. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-19-TST-006 - Failure containment. Inject purpose drift, overcollection, retention breach, inference leakage, deletion incompleteness. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-19-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-19-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-19-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-19-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.